

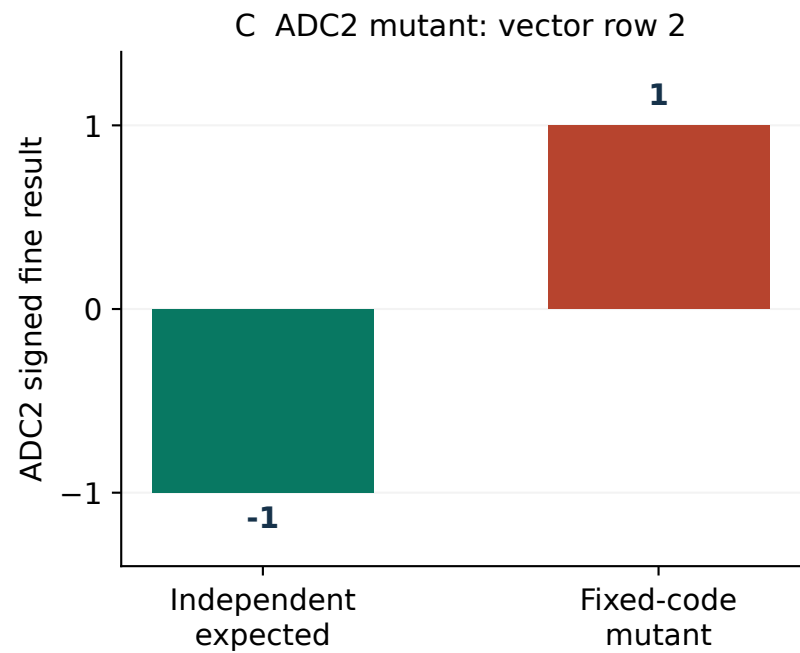
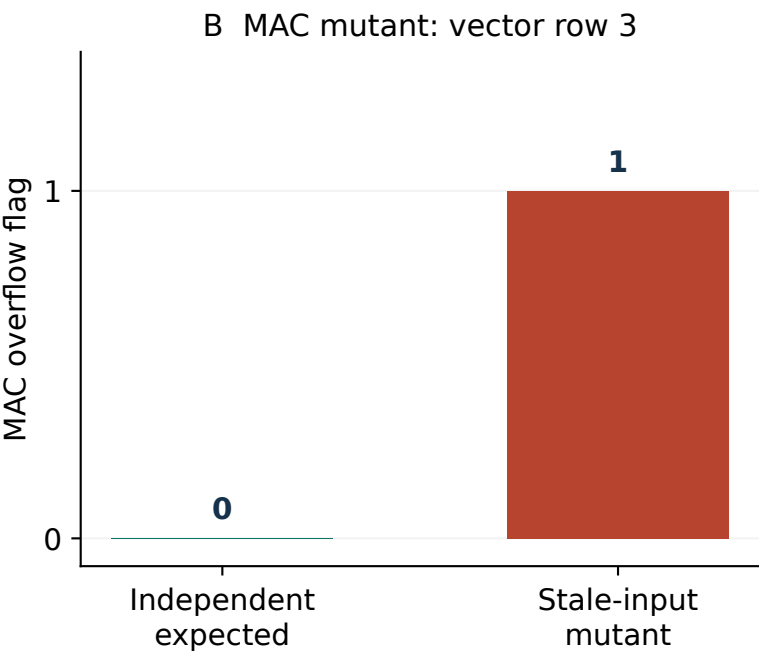
INDEPENDENT ARITHMETIC AUDIT

Local two-tool acceptance + deliberately stale-input negative controls

A Accepted simulations: exact raw-log counts

Recorded tool	MAC cases	ADC2 cases	Divider requests	Flag sequences
Verilator 5.020	40,010	40,000	10,048	8
Verilator 5.49 dev	40,010	40,000	10,048	8

MAC: 20,859 non-overflow cases are a subset of 40,010, not an extra count.
Divider: 9-cycle completion and ignored busy requests are explicitly logged.
Both tools replay the same vector hashes; counts are not summed into coverage.



Both intentional mutants fail: simulator return -6; audit runner return 1.
These are recorded assertion failures, not successful output samples.

8 runtime-path/environment unit tests also pass (process fixtures, not RTL simulation).

Source: docs/evidence/20260930/arithmetic_runner_ci/ | No reconstructed analog waveform.

Finite integer/protocol checks; no full-chip mapped, timing, SNDR or silicon claim.